

CY 508: Organometallic Chemistry

Course Code	CY 508
Course Name	Organometallic Chemistry
L-T-P-C	2-0-0-2
Prerequisites	CY 506 and CY 507
Intended for	M.Sc Chemistry (2nd year)
Distribution	core
Approval	9th Senate (May 2016)

Course Contents

- **Module-I** [6 Lectures] Introduction to organometallics: Brief history of organometallic chemistry, Werner Complexes, Types of Ligand, Soft Versus Hard Ligands, The Crystal Field, The Ligand Field, Back Bonding, Electroneutrality, Oxidation State, Coordination Number and Geometry, Effects of Complexation, Differences between Metals, Outer-Sphere Coordination, Kept Model, VBT, MOT, The Trans Effect and its theory
General Properties of Organometallic Complexes: The 18-Electron Rule, Limitations of the 18- Electron Rule, Electron Counting in Reactions
Metal Alkyls, Aryls, and Hydrides and Related π -Bonded Ligands Transition Metal Alkyls and Aryls, Related σ -Bonded Ligands, Metal Hydride Complexes, σ Complexes, Bond Strengths for Classical σ -Bonding Ligands
Carbonyls, Phosphine Complexes, and Ligand Substitution Reactions Metal Complexes of CO, RNC, CS and NO, phosphines and related ligands
- **Module-II** [10 Lectures]
Oxidative Addition and Reductive Elimination Concerted Additions, S_N2 Reactions, Reductive Elimination, Oxidative Coupling and Reductive Cleavage, Dissociative Substitution, Associative Mechanism, Redox Effects, the I Mechanism, and Rearrangements in Substitution, Photochemical Substitution, Steric and Solvent Effects in Substitution, Insertion and Elimination Reactions Involving CO, Insertions Involving Alkenes
Nucleophilic and Electrophilic Addition and Abstraction Nucleophilic Addition to CO, Electrophilic Addition, Electrophilic Abstraction of Alkyl Groups, Single-Electron Transfer Pathways
- **Module-III** [6 Lectures]
Homogeneous Catalysis Alkene Isomerization, Alkene Hydrogenation, Alkene Hydroformylation, Hydrocyanation of Butadiene, Alkene Hydrosilation and Hydroboration, Coupling Reactions, Surface and Supported Organometallic Catalysis, Grubbs and Schrock catalysts
Clusters and the Metal-Metal Bond Structures, The Isolobal Analogy, Synthesis, Reactions, Giant Clusters and Nanoparticles, Giant Molecules, Borane, carborane, metalocarborane, bimetallic and clusters complexes structure and application in catalysis
Applications of Organometallic Chemistry Alkene Metathesis, Dimerization, Oligomerization, and Polymerization of Alkenes, Activation of CO and CO₂, CH Activation, Organometallic Materials and Polymers, σ -Bond Metathesis
- **Module-IV** [6 Lectures]
Introduction to Bio-organometallic Chemistry : Organometallic enzymes and coenzymes, Vitamin B12 coenzyme, B12 model compounds, Organometallic compounds as drugs, Organometallic compounds as radiopharmaceuticals, tracers, ionophores and sensors
Introduction to metallocenes
Important Reactions :Beta hydride elimination, Olefin Metathesis and Cross Coupling Reactions, reaction with metals and transmetalation, hydrometallation, Heck reaction, Suzuki-Miyaura coupling, Sonogashira coupling, Stille coupling and Negishi coupling, industrial applications of cross coupling reactions.

Text Books

1. Robert H. Crabtree, **The Organometallic Chemistry of the Transition Metals**, 5th Edition, Wiley-Blackwell; 2009.
2. B D Gupta & A J Elias, Basic Organometallic Chemistry Univ. Press.
3. Christoph Elschenbroich, **Organometallics**
4. Dirk Steinborn, **Fundamentals of Organometallic**

References

1. R. C. Mehrota, **Organometallic Chemistry: A Unified Approach**, 2nd Edition New Age International Publishers, 2009
2. Whyman Robin, **Applied Organometallic Chemistry And Catalysis**, Oxford University Press, 2012.